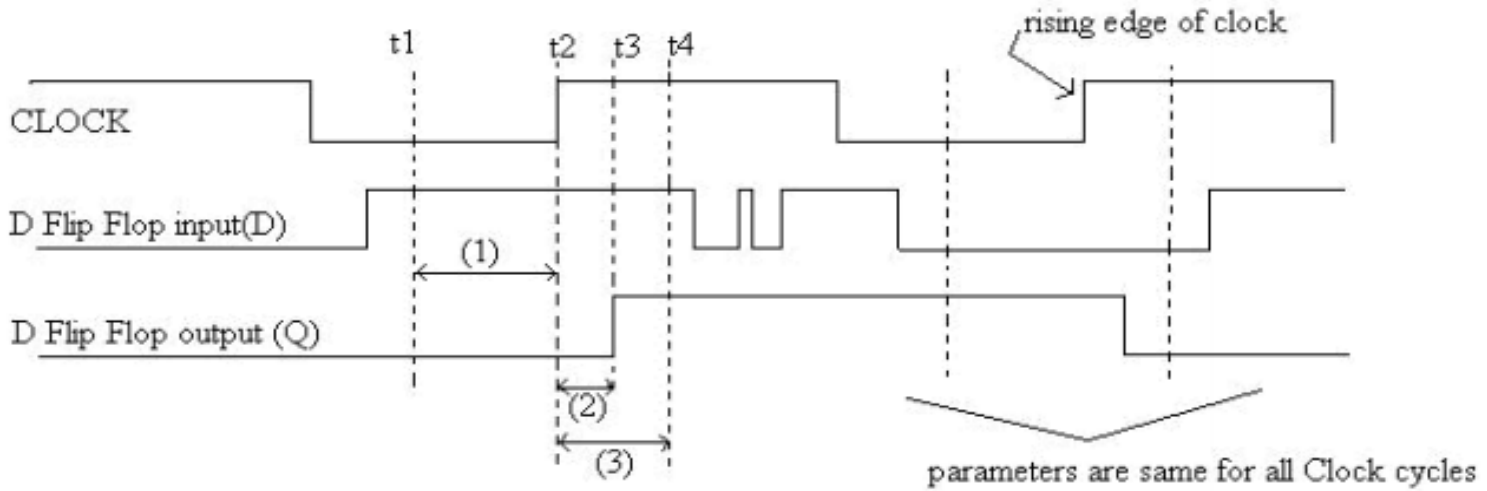
Setup and Hold Time



The timing diagram above illustrates three signals: the Clock, the Flip Flop Input (D) and the Flip Flop output (Q).

(1) is the Setup Time [t2 - t1]: the minimum amount of time Input must be held constant BEFORE the clock tick. Note that D is actually held constant for somewhat longer than the minimum amount. The extra "constant" time is sometimes called the setup margin.

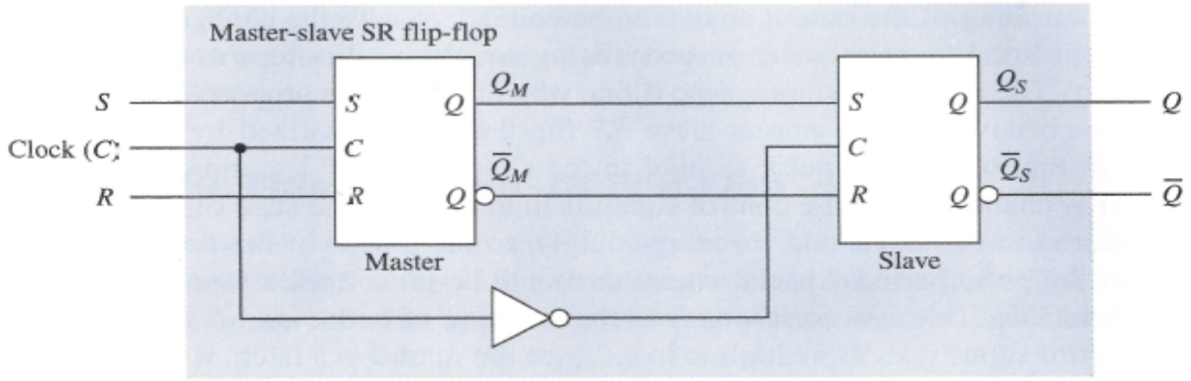
(2) is the Propagation delay of the Flip Flop [t3 - t2]: this is the time that it takes for the new input to be to propagate and influence the output.

(3) is the Hold time [t4 - t2]: the minimum amount of time the Input is held constant AFTER the clock tick. Note that Q is actually held constant for somewhat longer than the minimum amount. The extra "constant" time is sometimes called the hold margin.

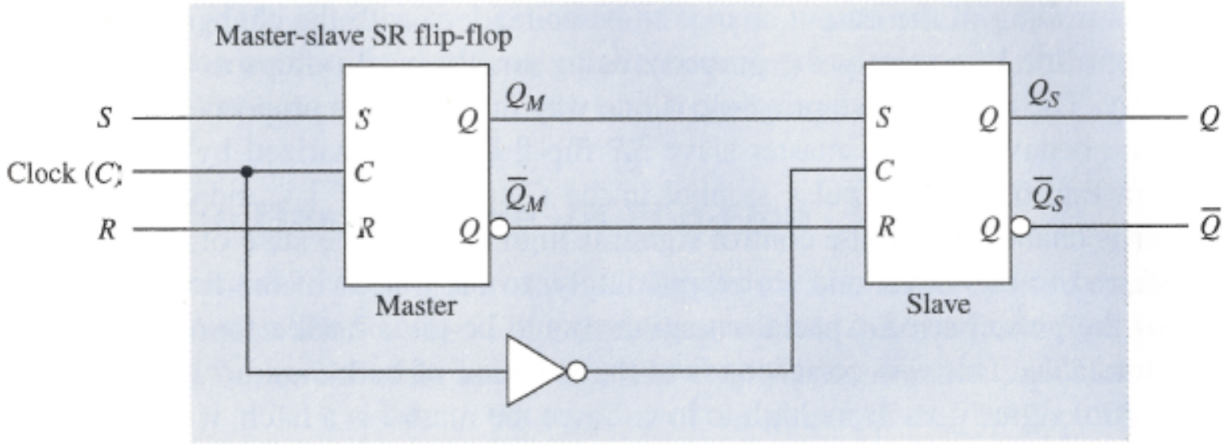
(The above timing diagram has 2 clock cycles; the timing parameters for the second cycle will also be similar to that of the first cycle)

Master-Slave SR Flip-Flop

- Two sections, each capable of storing a binary symbol.
- First section is referred to as the master and the second section as the slave.
- Information is entered into the master on one edge or level of a control signal and is transferred to the slave on the next edge or level of the control signal.
- Each section is a latch.

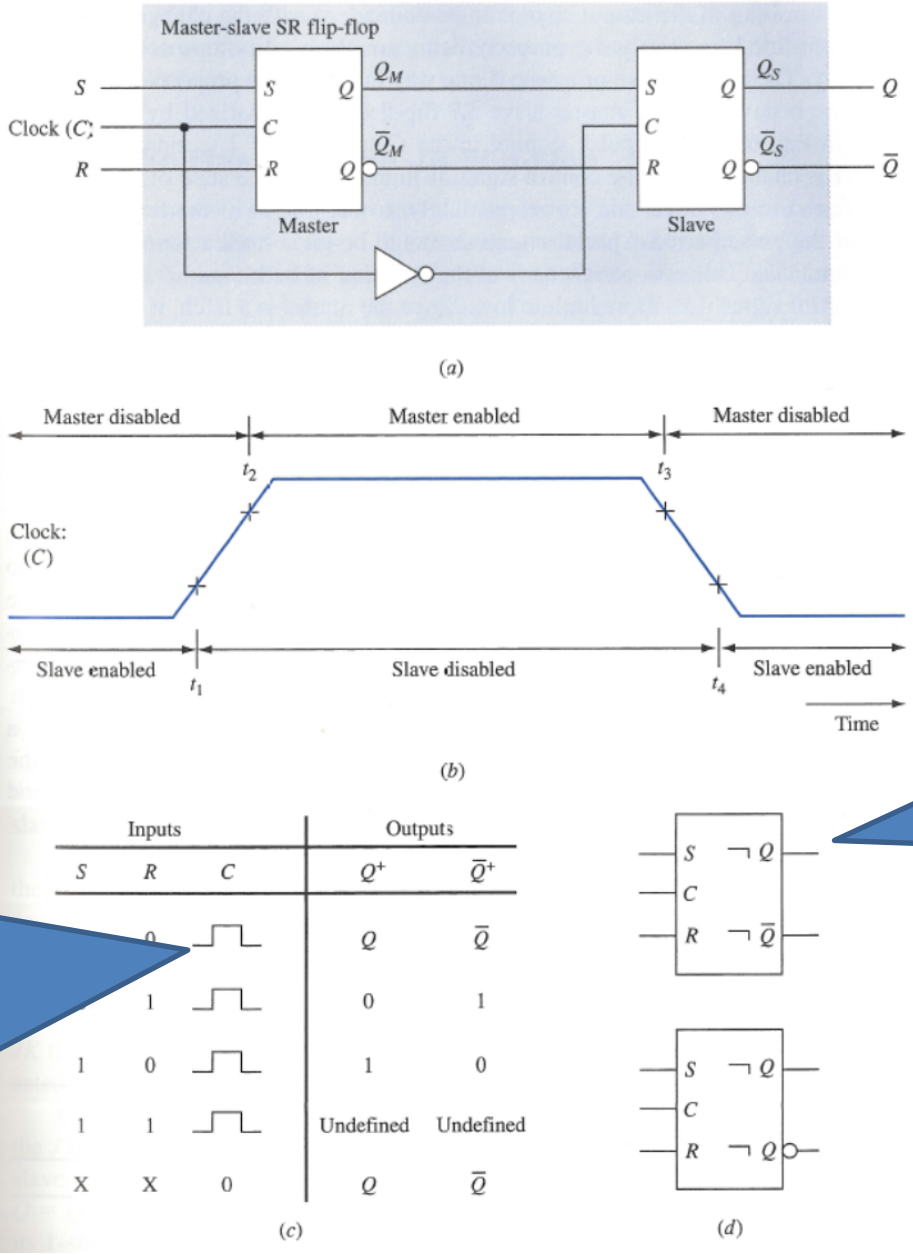


Master-Slave SR Flip-Flop



- C = 0:
  - Master is disabled. Any changes to S,R ignored.
  - Slave is enabled. Is in the same state as the master.
- C = 1:
  - Slave is disabled (retains state of master)
  - Master is enabled, responds to inputs. Changes in state of master are not reflected in disabled slave.
- C = 0:
  - Master is disabled.
  - Slave is enabled and takes on new state of the master.
- Important: For short periods during rising and falling edges, both master and slave are disabled.

Master-Slave SR Flip-Flop



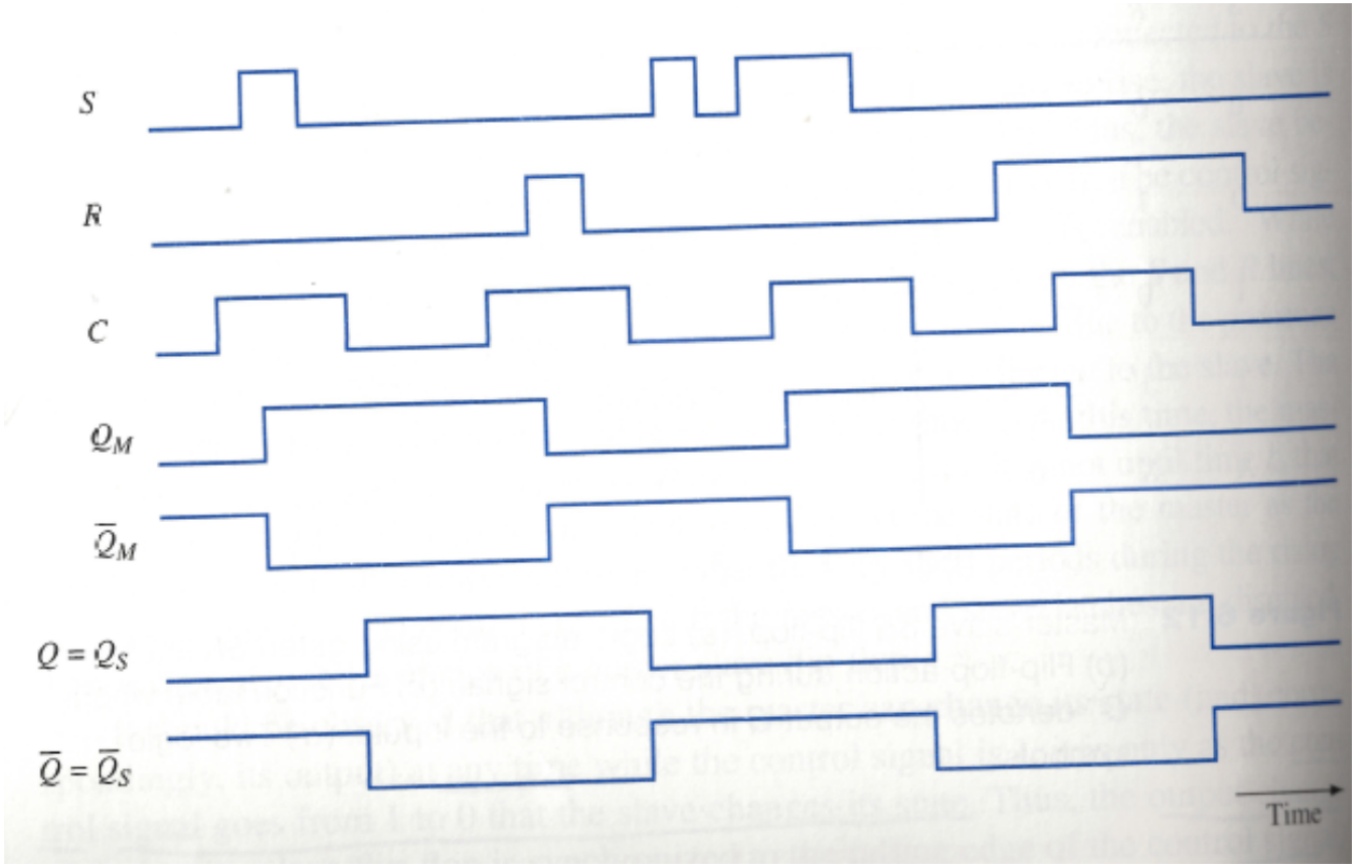
symbol indicates master enabled when C = 1 and state of master transferred to slave at the end of the pulse

Slave only takes on state of the master at t_4.

Postponed output indicator: output change postponed until end of pulse

control signal goes from high to low we are in an unpredicable state. Can cause

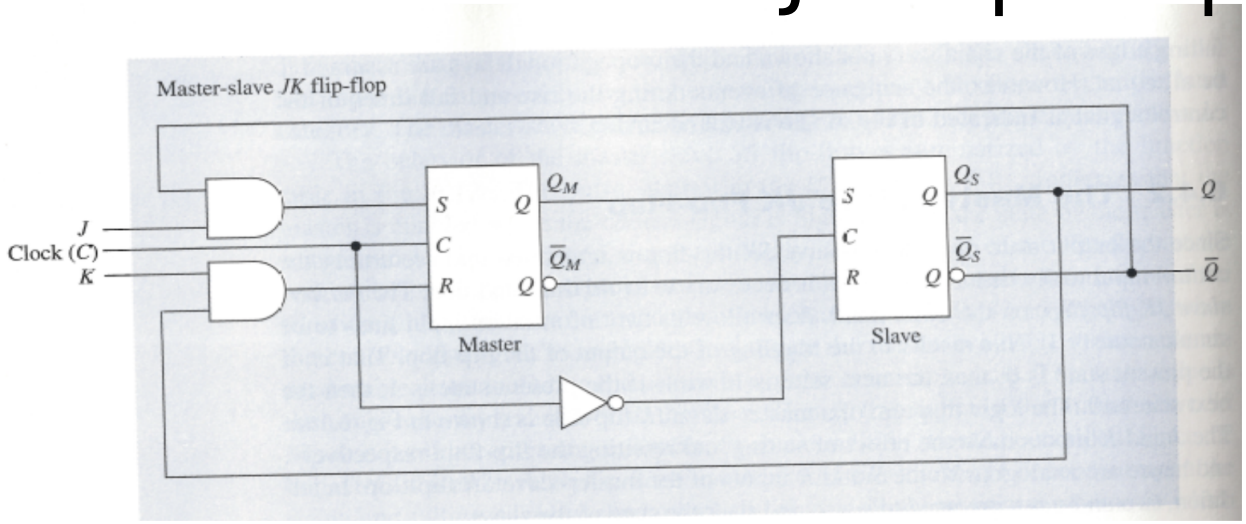
Timing Diagram for Master-Slave SR flip-flop



Master-Slave JK Flip-Flop

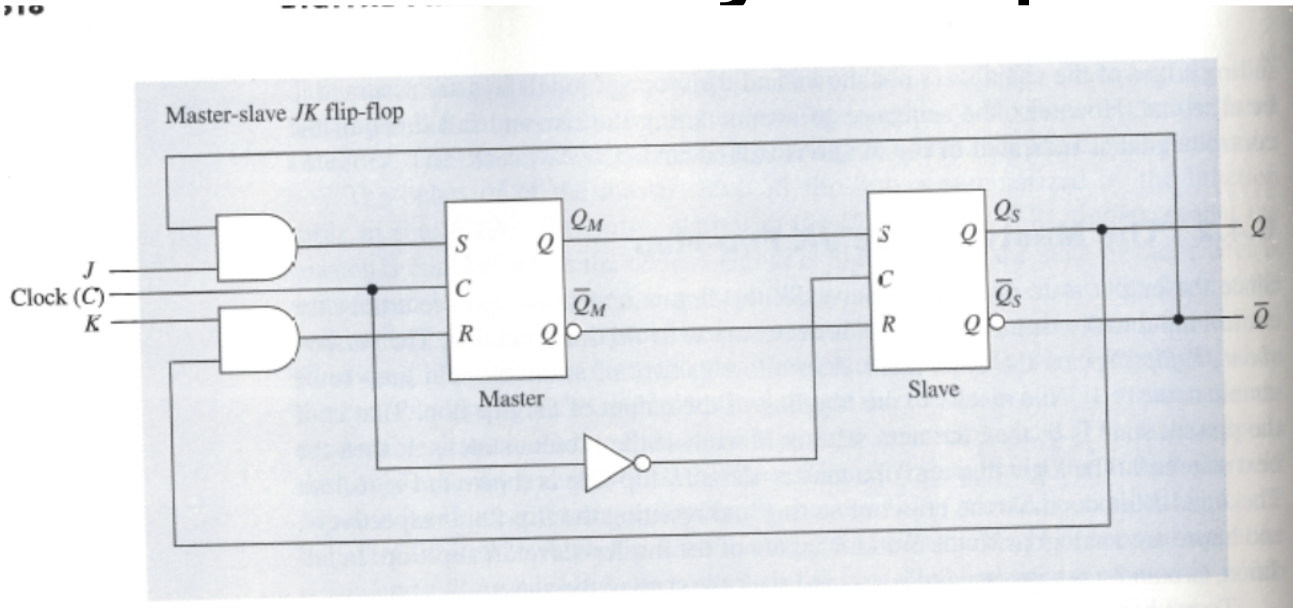
- The output state of a master-slave SR flip-flop is undefined upon returning the control input to 0 when S = R = 1.
  - Necessary to avoid this condition.
- Master-slave JK flip-flop allows its two information input lines to be simultaneously 1.
  - Results in toggling the output of the flip flop.

Master-Slave JK Flip-Flop



- Assume in 1-state, C = 0, J = K = 1.
  - Due to feedback, the output of the J-gate is 0, output of K-gate is 1.
  - If clock is changed to C = 1 then master is reset.
- Assume in 0-state, C = 0, J = K = 1.
  - Due to feedback, the output of the J-gate is 1, output of K-gate is 0.
  - If clock is changed to C = 1 then master is set.
- 1 on J input line, 0 on K input line sets the flip-flop.
  - If in 1-state, unchanged b/c S,R set to 0.
  - If in 0-state, S set to 1, R set to 0.
- 0 on J input, 1 on K input line resets the flip-flop. Why?

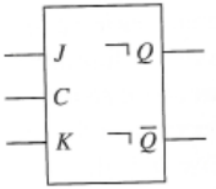
Master-Slave JK Flip-Flop

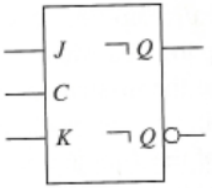


(a)

Inputs			Outputs	
J	K	C	Q^+	$\overline{Q}^+$
0	0	⎍	Q	$\overline{Q}$
0	1	⎍	0	1
1	0	⎍	1	0
1	1	⎍	$\overline{Q}$	Q
X	X	0	Q	$\overline{Q}$

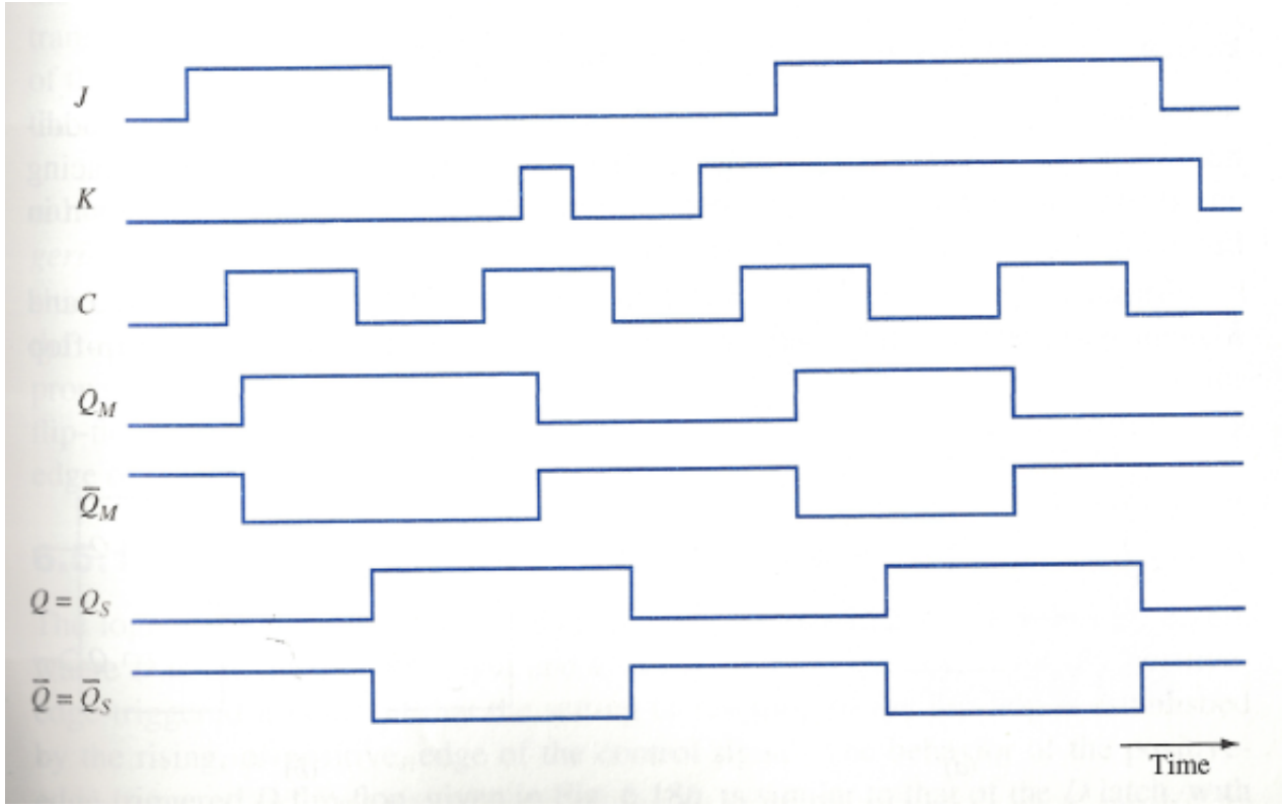
(b)



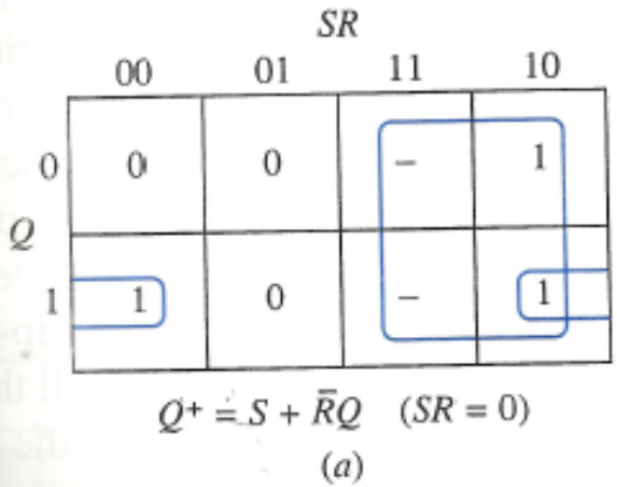


(c)

Timing Diagram for Master-Slave JK Flip-Flop



Characteristic Equations



$Q^+ = S + \bar{R}Q \quad (SR = 0)$

(*a*)

Flip-flop type	Characteristic equation
SR	$Q^+ = S + \bar{R}Q \quad (SR = 0)$
JK	$Q^+ = J\bar{Q} + \bar{K}Q$
D	$Q^+ = D$
T	$Q^+ = T\bar{Q} + \bar{T}Q = T \oplus Q$

(*b*)

Figure 6.25 Characteristic equations. (*a*) Derivation of characteristic equation for an *SR* flip-flop. (*b*) Summary of characteristic equations.

Registers

- A collection of flip-flops taken as an entity.
- Function: Hold information within a digital system so that it is available to the logic elements during the computing process.
- Each combination of stored information is known as the state or content of the register.
- Shift register: Registers that are capable of moving information upon the occurrence of a clock-signal.
  - Unidirectional
  - bidirectional

Registers

- Two basic ways in which information can be entered/outputted
  - Parallel: All 0/1 symbols handled simultaneously. Require as many lines as symbols being transferred.
  - Serial: Involves the symbol-by-symbol availability of information in a time sequence.
- Four possible ways registers can transfer information:
  - Serial-in/serial-out
  - Serial-in/parallel-out
  - Parallel-in/parallel-out
  - Parallel-in/serial-out